

Project Title : Smart Medical Insurance Cost Predictor with NGO & Welfare Aid Guidance

DS Project Review-2



Under the Guidance of

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Outline of Project Work

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- Introduction

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- Research Objectives

- Conclusion & Future Work

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Problem Statement

- Rising and unpredictable healthcare costs make it hard for individuals to plan financially, especially for high-risk profiles (older age, high BMI, smokers).
- Existing insurance-cost prediction tools stop at generating an estimate — they don't tell financially vulnerable patients what support exists for them.
- Many patients eligible for NGO, charitable, or government welfare schemes never discover them, because such information is scattered and not linked to their actual financial situation.
- There is no unified system that combines cost estimation with personalized, actionable guidance on financial assistance options.

Objectives

- Perform structured data cleaning, transformation, and EDA on the Medical Cost Personal Dataset to surface patterns among cost-driving factors (age, BMI, smoking, region, etc.).
- Encode categorical attributes and build/compare Linear Regression, Random Forest Regression, and Gradient Boosting Regression models, evaluated via MAE, RMSE, and R^2 .
- Curate a supporting dataset of NGOs, welfare organizations, and government schemes with eligibility rules (income thresholds, condition type, region, aid ceiling).
- Implement a rule-based matching engine linking predicted cost + user profile to eligible aid organizations and likely support amount.
- Build an interactive dashboard integrating cost prediction, aid-matching results, and visual analytics (feature importance, model comparison).

Literature Survey

Authors	Title of the Paper	Methodology	Inferences	Limitation/Future work
Orji et al., 2024 (arXiv)	Machine Learning For An Explainable-Cost Prediction of Medical Insurance	Ensemble regression (XGBoost, GBM, Random Forest) + SHAP/ICE for explainability, on 986-record Kaggle dataset	Tree-based ensembles predict charges well; SHAP/ICE show smoking, BMI, age as dominant cost drivers	Focuses only on prediction/explainability — no downstream support for patients who can't afford the estimate
IJRASET, 2023	Medical Insurance Cost Prediction using Machine Learning	Linear Regression, Decision Tree Regression, Gradient Boosting Regression, deployed via Streamlit on a Kaggle medical cost dataset	Gradient Boosting/tree-based regressors outperform plain linear regression; R^2 used for comparison	Outputs a single cost estimate only — no guidance for financially struggling patients
Hong et al., 2025 (PMC/NCBI)	Predicting a Need for Financial Assistance in Emergency Department Care	Logistic Regression, Decision Tree, Random Forest on 1M+ ED billing records to predict bill non-payment likelihood	Decision Tree correctly flagged 87% of patients likely needing financial assistance	Identifies <i>need</i> for aid but doesn't recommend or match specific organizations/schemes

Problem Gap Identification

- Cost-prediction research (regression/ensemble ML) stops at the number — none connect the prediction to actionable relief.
- Financial-assistance-need studies (e.g., ED billing prediction) flag *who* needs help but don't say *which* NGO/scheme could help or *how much*.
- No existing dataset or system links individual cost prediction to a structured, eligibility-matched directory of welfare/NGO support.
- **Gap:** absence of an integrated cost-prediction + rule-based-aid-matching pipeline that turns a raw estimate into a personalized next step.

Innovation, Creativity, and Novelty

- Combines two normally separate domains — insurance cost prediction (ML regression) and welfare/NGO aid eligibility (rule-based matching) — into one pipeline.
- Introduces a self-curated NGO/government-scheme dataset (eligibility rules by income, condition, region, aid ceiling) — a genuinely new data asset, since no public dataset links medical costs to assistance programs.
- Matching engine is deliberately rule-based, not ML, for transparency — a black-box recommendation is inappropriate when advising vulnerable users on financial aid.
- Dashboard tells users not just "what you'll pay" but "who can help and how much" — a predictive tool becomes an actionable support tool.
- Requires two coordinated data-cleaning pipelines (numeric health dataset + curated rules dataset), going beyond standard single-dataset course projects.

Feasibility and Project Plan

Phase	Duration	Deliverable
1	Week 1-2	Data collection & cleaning — Medical Cost dataset + curating NGO/welfare dataset
2	Week 3	EDA — distributions, correlations, smoking/BMI/age impact
3	Week 4	Model development — Linear Regression, Random Forest, Gradient Boosting
4	Week 5	Model evaluation — MAE/RMSE/R ² comparison, best-model selection
5	Week 6	Rule-based aid-matching engine built from NGO dataset
6	Week 7	Dashboard integration — prediction + matching + analytics
7	Week 8	Functional testing, report writing, final presentation
8	Week 9	Testing, refinement, documentation